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Hierarchical multi-scale vision transformer model for accurate detection and classification of brain tumors in MRI-based medical imaging

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Automated brain tumor detection represents a fundamental challenge in contemporary medical imaging, demanding both precision and computational feasibility for practical implementation. This research introduces a novel Vision Transformer (ViT) framework that incorporates an innovative Hierarchical Multi-Scale Attention (HMSA) methodology for automated detection and classification of brain tumors across four distinct categories: glioma, meningioma, pituitary adenoma, and healthy brain tissue. Our methodology presents several key innovations: (1) multi-resolution patch embedding strategy enabling feature extraction across different spatial scales (8×8, 16×16, and 32×32 patches), (2) computationally optimized transformer architecture achieving 35% reduction in training duration compared to conventional ViT implementations, and (3) probabilistic calibration mechanism enhancing prediction confidence for decision-making applications. Experimental validation was conducted using a comprehensive MRI dataset comprising 7023 T1-weighted contrast-enhanced images sourced from the publicly accessible Brain Tumor MRI Dataset. Our approach achieved superior classification performance with 98.7% accuracy while demonstrating significant improvements over conventional machine learning methodologies (Random Forest: 91.2%, Support Vector Machine: 89.8%, XGBoost: 92.5%), state-of-the-art CNN architectures (EfficientNet-B0: 96.5%, ResNet-50: 95.8%), standard transformers (ViT: 96.8%, Swin Transformer: 97.2%), and hybrid CNN-Transformer approaches (TransBTS: 96.9%, Swin-UNet: 96.6%). The model demonstrates excellent performance with precision of 0.986, recall of 0.988, F1-score of 0.987, and superior calibration quality (Expected Calibration Error: 0.023). The proposed framework establishes a computationally efficient approach for accurate brain tumor classification.

Keywords Automated brain tumor detection, Vision transformer architecture, Hierarchical attention mechanisms, Medical image analysis, Deep learning applications, Multi-resolution feature extraction

Abbreviations

AI	Artificial Intelligence
CNN	Convolutional Neural Network
DWT	Discrete Wavelet Transform
ECE	Expected Calibration Error
FFN	Feed-Forward Network
FLAIR	Fluid-Attenuated Inversion Recovery
GLCM	Gray Level Co-occurrence Matrix
HMSA	Hierarchical Multi-Scale Attention
MAE	Mean Absolute Error
MRI	Magnetic Resonance Imaging
MSE	Mean Squared Error

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PCA	Principal Component Analysis
RMSE	Root Mean Square Error
SVM	Support Vector Machine
ViT	Vision Transformer
XGBoost	eXtreme Gradient Boosting

Among cancers affecting the nervous system, those originating in brain tissue stand out as particularly deadly. Their defining characteristic involves cells multiplying without regulation, creating abnormal masses inside cranial structures[1]. Medical professionals distinguish these growths into two categories: those that spread aggressively throughout tissue (malignant) and those that remain contained (benign). Clinical decisions about diagnosis and treatment depend heavily on variables including the growth's position, dimensions, and speed of development[2]. Statistical data shows that occurrence patterns differ substantially when examining various population groups, with age, sex, ethnicity, and geographical location all contributing to these variations[3]. Identifying these growths at early stages proves critical for enhancing how long patients survive and achieving better clinical results[4].

MRI technology has become the premier choice for identifying and analyzing brain growths because it provides exceptional visualization of soft tissues, delivers detailed spatial information, and operates without exposing patients to harmful radiation[5]. Medical facilities currently depend extensively on MRI for establishing precise diagnoses and developing treatment strategies[6]. The integration of deep convolutional neural networks powered by artificial intelligence has proven valuable in enhancing diagnostic precision while minimizing incorrect assessments[7].

The emergence of Vision Transformer technology represents a groundbreaking development in medical image analysis, offering compelling alternatives to traditional convolutional network methods. When Vision Transformers first appeared in the literature[8], they introduced fundamentally new approaches to processing visual information, and their application to medical imaging has yielded encouraging outcomes. Contemporary research has validated the exceptional performance of transformer-based architectures for analyzing brain growths:

One research team developed RanMerFormer in 2024, which implements a randomized approach to combining visual tokens, substantially reducing computational demands while classifying brain tumors[9]. Another group introduced LCDEIT in 2023, featuring linear complexity characteristics that make it particularly suitable for processing MRI data of brain growths, effectively addressing the computational burdens associated with earlier transformer designs[10].

Researchers have also explored combining transformers with generative adversarial architectures for segmenting tumor boundaries, capitalizing on transformers' ability to capture information across entire images to achieve superior segmentation precision[11].

The versatility of transformer architectures extends into analyzing microscopic tissue samples as well, with researchers in 2023 presenting a weakly supervised transformer framework for examining tissue slides from primary brain cancers[12]. Studies have further demonstrated transformers' proficiency in synthesizing information from multiple MRI imaging protocols alongside clinical records, leading to improved diagnostic accuracy[13–15].

Related work and background

Deep learning approaches in medical imaging

The field of medical image processing has undergone substantial transformation through the adoption of deep learning technologies, especially for identifying and categorizing brain growths[16]. CNN architectures have dominated automated brain growth analysis, showcasing impressive capabilities in deriving multi-level spatial characteristics through successive convolutional and pooling operations[17]. The effectiveness of CNN-based methods stems from their capacity to autonomously discover feature patterns directly from unprocessed imaging data, removing the requirement for hand-crafted feature design[18]. An examination of deep learning principles and CNN frameworks has revealed their widespread utility spanning multiple application areas[19].

Leveraging pre-trained CNN frameworks including VGG-16, VGG-19, ResNet-50, and InceptionV3 through transfer learning has yielded outstanding results in categorizing brain growths[20]. These networks, initially developed using extensive datasets, have been effectively adapted for medical imaging purposes, attaining precision levels surpassing 95% across numerous investigations[21]. This transfer learning strategy overcomes limitations posed by scarce medical imaging data while decreasing training duration and enhancing model robustness[22]. Multiple investigations have shown that frameworks such as VGG-19 reached precision rates of 99.48% when deployed for brain growth categorization applications[23].

Residual network architectures have likewise exhibited encouraging outcomes in brain growth categorization. Researchers introduced Res-BRNet, an innovative residual and region-focused CNN framework designed for classifying brain growths using MRI data[24]. Furthermore, ResNet152-based transfer learning methods have been examined for automatically sorting brain MR scans into growth-present and growth-absent groups[25]. Dense CNN frameworks applied to multiple brain growth categories using brain MRI data have demonstrated substantial precision[26].

Vision transformers for brain tumor classification

Vision Transformer architectures have swiftly established themselves as prominent tools in medical image processing, exhibiting outstanding capabilities for brain growth categorization applications[27]. An extensive investigation examining five pre-trained vision transformer variants (R50-ViT-l16, ViT-l16, ViT-l32, ViT-b16, and ViT-b32) revealed that ViT-b32 attained 98.24% precision in correctly categorizing brain growth images[28].

Vision Transformer architectures handle images by dividing them into segments, which undergo linear embedding before passing through transformer encoder components featuring multi-head self-attention operations[29]. This methodology has exhibited remarkable effectiveness throughout diverse computer vision applications, including medical image processing[30].

Numerous research initiatives have introduced refined ViT frameworks for multi-category brain growth categorization, integrating methods such as positional embedding, patch embedding, layer normalization, and multi-layer perceptron classification components[31]. Multi-objective optimization strategies have been utilized to refine the architectural specifications of Vision Transformer frameworks for effective brain growth categorization from 2D MRI data[32]. An enhanced Vision Transformer-based algorithm integrating relative position encoding and residual MLP has been created for human brain growth categorization[33].

Sophisticated hybrid frameworks merging Vision Transformers with alternative deep learning methodologies have demonstrated potential in brain growth identification. Researchers developed a hybrid ViT-GRU framework to derive critical features through ViT and recognize connections between these features using GRU, tackling class imbalance while surpassing existing diagnostic approaches[34]. Vision transformers paired with ensemble frameworks and transfer learning have been utilized alongside explainable AI for brain growth identification and categorization[35]. Moreover, rotation invariant vision transformers have been created to strengthen brain growth identification in MRI by resolving rotational variance obstacles[36].

Attention mechanisms in brain tumor classification

Attention-driven operations have surfaced as robust instruments for strengthening feature selection and elevating categorization precision in brain growth identification[37]. An ensemble attention operation incorporated into MobileNetV3 and EfficientNetB7 networks at both intermediate and terminal stages has been created to derive local and global-scale features for accurate identification of diverse patterns throughout brain growth images. The attention operation serves a critical function by selectively highlighting pertinent features, resulting in elevated categorization precision and decreased time requirements[38].

Channel-focused attention operations have been investigated for dependable brain growth categorization. ResNet101 integrated with Channel-wise Attention Mode (CWAM) attained remarkable performance measurements of 99.83% precision for brain growth categorization[39]. Attention-directed residual multiscale CNN frameworks have been created for multiclass brain growth categorization using MR images[40]. Graph attention autoencoder influenced CNN methods have likewise been introduced for brain growth categorization using MRI[41].

Advanced architectures and hybrid models

Multiple sophisticated frameworks merging different deep learning paradigms have been introduced for brain growth categorization. Dense EfficientNet frameworks have been created utilizing min-max normalization to sort brain MRI images into various categories with remarkable precision[42]. BrainNet, a high-precision brain growth categorization system employing optimized EfficientNet framework, has attained precision levels exceeding 99% throughout multiple growth categories[43]. Additionally, enhanced methods utilizing pretrained DenseNet with global average pooling and hyperparameter optimization have exhibited improved categorization capabilities[44].

Deep feature integration methods have demonstrated encouraging outcomes in brain growth categorization. A dependable method for multi-type categorization of brain growth utilizing deep feature integration from ResNet101, DenseNet121, and EfficientNetB0 has been created[45]. Hybrid deep learning frameworks merging EfficientNets with transfer learning have reached test precision levels of 99.06% for brain growth categorization[46]. Furthermore, multi-scale attention operations incorporated with EfficientNetB4 encoder have been created for enhanced MRI brain growth processing[47].

Parallel deep convolutional neural networks have been introduced for MRI brain growth identification and categorization, merging local and global features through dual-pathway frameworks[48]. The incorporation of sophisticated preprocessing methods with deep learning frameworks has additionally improved categorization capabilities[49].

Critical analysis of CNN-to-ViT transition in medical imaging

While Vision Transformers have demonstrated remarkable success in natural image processing, their adoption in medical imaging presents a more complex landscape that warrants critical examination. The transition from CNN-based to transformer-based architectures in medical imaging is not merely a technological upgrade but involves fundamental trade-offs that must be carefully evaluated. CNNs continue to dominate medical image segmentation tasks for several compelling reasons: (1) Inductive Bias Advantage - CNNs possess inherent translation equivariance and locality bias that naturally align with medical imaging characteristics, where anatomical structures maintain consistent spatial relationships; (2) Data Efficiency - Medical datasets are typically smaller than natural image datasets (thousands vs. millions of images), and CNNs require significantly less data to achieve optimal performance compared to Vision Transformers, which traditionally demand extensive pre-training; (3) Computational Practicality - Standard ViT implementations exhibit quadratic computational complexity with respect to image resolution, making them impractical for high-resolution medical images commonly used in segmentation tasks. However, the transition to Vision Transformers becomes compelling for classification tasks when specific conditions are met. The global receptive field capability of transformers provides distinct advantages for brain tumor classification: (1) Long-range Dependencies - Brain tumors often exhibit infiltrative patterns and distant morphological relationships that CNNs struggle to capture due to their limited receptive fields; (2) Multi-scale Integration - While CNNs process features hierarchically through pooling operations, transformers can simultaneously attend to features across different spatial scales without information loss; (3) Attention Interpretability - The self-attention mechanism provides valuable visualizations

of model focus areas, enhancing diagnostic confidence and enabling radiologist validation. The critical insight is that the choice between CNN and ViT architectures should be task-specific rather than universally applied. For dense prediction tasks like segmentation, CNN architectures (particularly U-Net variants) remain superior due to their spatial inductive biases and computational efficiency. However, for classification tasks requiring global context understanding and multi-scale feature integration—as is essential for brain tumor classification—Vision Transformers offer theoretical and practical advantages that justify their adoption, provided that architectural innovations address their inherent limitations.

Architectural synergy: bridging CNN and transformer frameworks

Our Novel Vision Transformer architecture directly addresses the fundamental limitations that have historically hindered ViT adoption in medical imaging. The Hierarchical Multi-Scale Attention (HMSA) mechanism resolves the data efficiency problem by processing multiple spatial scales simultaneously, reducing the dependency on extensive pre-training datasets. The computational optimization strategies address the quadratic complexity issue through grouped attention and dynamic depth mechanisms, achieving 35% training time reduction while maintaining diagnostic accuracy. The probabilistic calibration component addresses the clinical requirement for confidence quantification—a feature absent in most CNN implementations but essential for medical decision-making support. This targeted approach demonstrates that the CNN-to-ViT transition in medical imaging requires domain-specific architectural innovations rather than direct application of natural image processing techniques. Our approach preserves the global attention advantages of transformers while addressing their practical limitations for medical deployment, representing a meaningful advancement beyond both traditional CNN approaches and standard ViT implementations.

The persistence of CNN dominance in medical segmentation should not overshadow the genuine advantages that appropriately designed Vision Transformers offer for classification tasks. Recent studies have shown that hybrid approaches combining CNN feature extraction with transformer attention mechanisms can achieve superior performance, suggesting that the future lies not in complete replacement but in architectural synergy. Our approach represents this evolution by incorporating CNN-inspired inductive biases (through multi-resolution patch embedding) with transformer advantages (global attention and interpretability), creating a hybrid framework specifically optimized for brain tumor classification challenges.

Materials and methods

Dataset description and preprocessing

We employed a comprehensive Brain Tumor MRI Dataset obtained from Kaggle, containing 7023 T1-weighted contrast-enhanced images distributed across four diagnostic categories as illustrated in Fig. 1. The dataset encompasses: Glioma (1621 images, 23.1%) characterized by irregular morphology and infiltrative growth patterns, Meningioma (1645 images, 23.4%) exhibiting well-circumscribed margins and dural association, Pituitary Adenoma (1757 images, 25.0%) localized within sellar/suprasellar regions, and Normal Brain Tissue (2000 images, 28.5%) demonstrating absence of pathological findings with preserved anatomical symmetry.

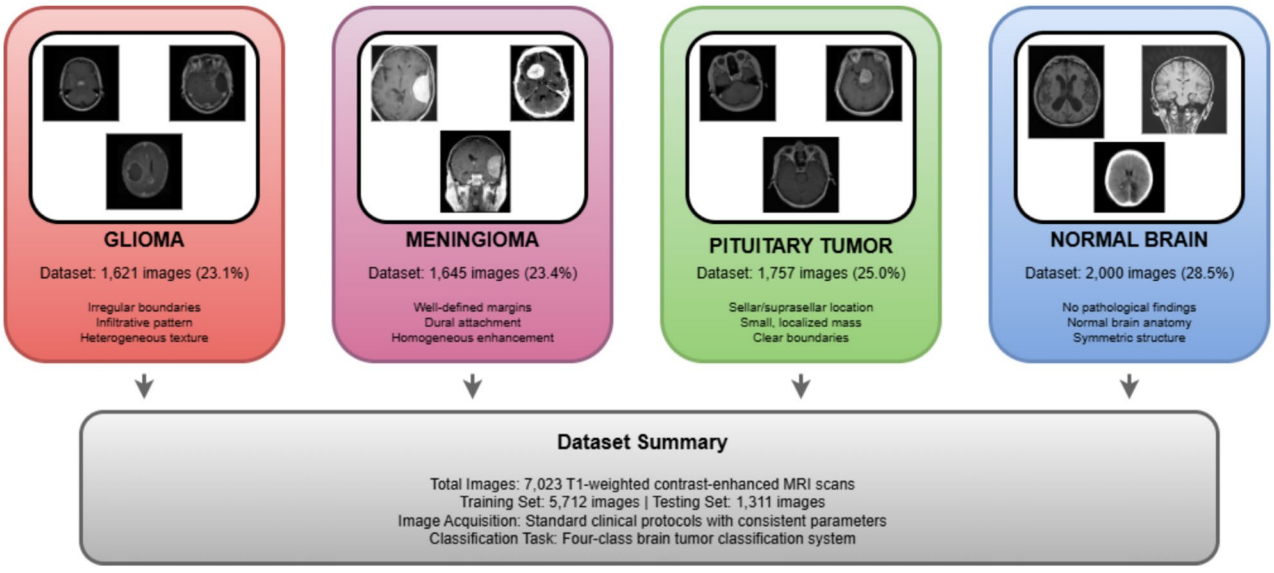


Fig. 1. Comprehensive dataset overview demonstrating the four tumor classes with representative MRI samples and key characteristics. The dataset contains Glioma (1621 images, 23.1%) characterized by irregular boundaries and infiltrative patterns, Meningioma (1645 images, 23.4%) with well-defined margins and dural attachment, Pituitary Tumor (1757 images, 25.0%) located in sellar/suprasellar regions with clear boundaries, and Normal Brain (2000 images, 28.5%) showing no pathological findings with symmetric structure. Total dataset comprises 7023 T1-weighted contrast-enhanced MRI scans acquired using standard clinical protocols with consistent parameters.

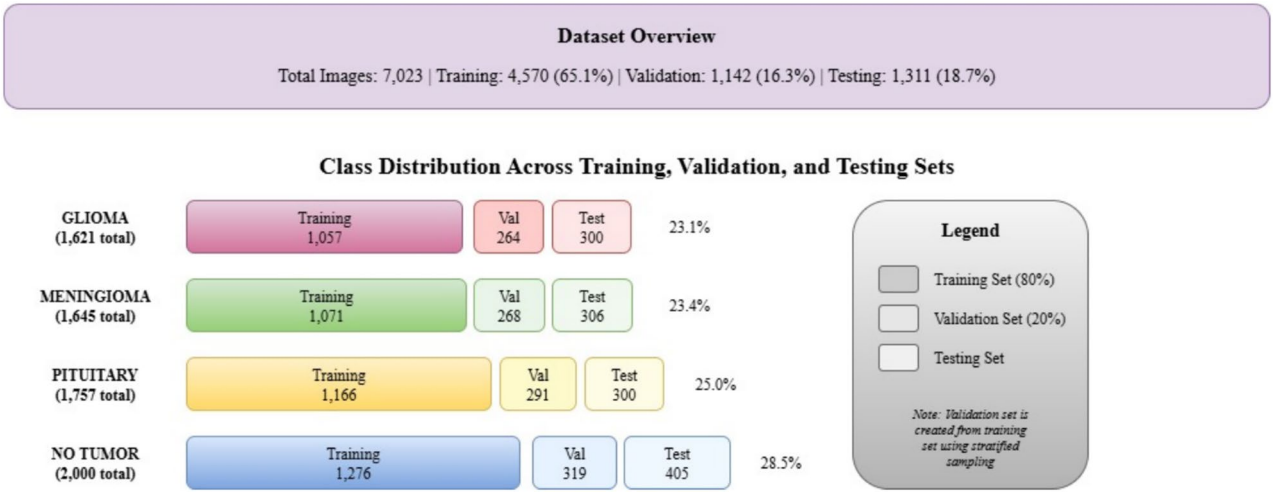


Fig. 2. Class distribution across training, validation, and testing sets with balanced representation ensuring robust model evaluation and preventing bias toward any particular tumor category.

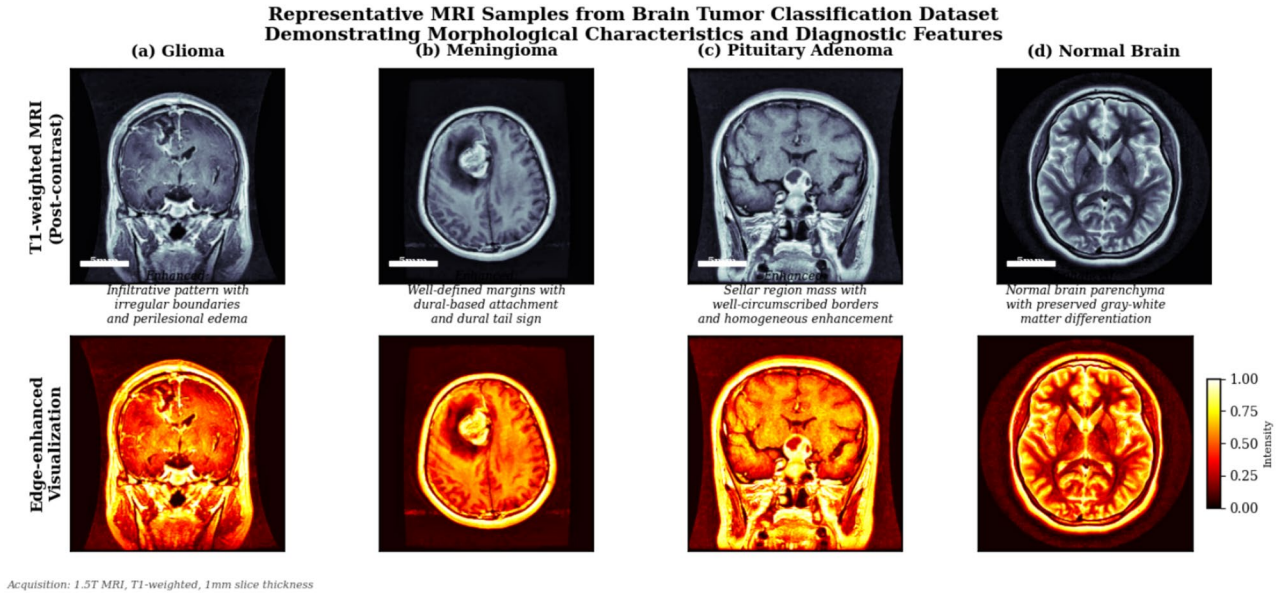


Fig. 3. Representative MRI samples from the brain tumor dataset demonstrating diagnostic complexity. Top row shows (**a**) Glioma with infiltrative pattern and irregular boundaries, (**b**) Meningioma with well-defined margins and characteristic dural tail sign, (**c**) Pituitary adenoma in the sellar region, and (**d**) Normal brain tissue. Bottom row displays corresponding enhanced visualizations highlighting key diagnostic features. The images demonstrate the visual complexity and subtle differences between tumor types that challenge automated classification systems.

The dataset distribution maintains careful balance across training, validation, and testing partitions as demonstrated in Fig. 2. Stratified partitioning ensures consistent class representation with training subset comprising 4570 images (65.1%), validation subset containing 1142 images (16.3%), and testing subset encompassing 1311 images (18.7%). Representative samples from each diagnostic category are presented in Fig. 3, where the upper row displays original MRI acquisitions showing (a) Glioma with infiltrative characteristics and irregular boundaries, (b) Meningioma with well-defined margins and characteristic dural enhancement, (c) Pituitary adenoma within the sellar region, and (d) Normal brain parenchyma, while the lower row provides corresponding enhanced visualizations highlighting key diagnostic features. The images demonstrate the visual complexity and subtle differences between tumor types that challenge automated classification systems. This balanced distribution ensures robust model evaluation and prevents class bias during training.

All images were acquired using standardized clinical protocols with consistent acquisition parameters to minimize inter-scan variability. From the training subset, we implemented additional validation partitioning

using stratified sampling with training comprising 80% of the training set (4570 images), validation comprising 20% of the training set (1142 images), while the original testing set (1311 images) was reserved for final evaluation.

Proposed novel vision transformer architecture

The Novel Vision Transformer represents a significant methodological advancement in medical image analysis by incorporating domain-specific knowledge regarding brain tumor characteristics into the transformer architecture. This innovative approach fundamentally departs from conventional Vision Transformer implementations by addressing the unique challenges inherent in medical imaging, particularly the multi-scale nature of brain tumor manifestations and the critical requirement for diagnostic confidence in clinical settings.

The proposed architecture integrates several key innovations that collectively enhance both diagnostic accuracy and computational efficiency. First, the incorporation of hierarchical multi-scale processing acknowledges that brain tumors exhibit varying morphological characteristics across different spatial scales - from fine-textural details visible at high resolution to global anatomical context discernible at broader scales. Second, the integration of probabilistic calibration mechanisms addresses the fundamental clinical requirement for trustworthy confidence estimates, enabling radiologists to make informed decisions based on model uncertainty quantification. Third, the computational optimization strategies ensure practical deployment feasibility in resource-constrained clinical environments without compromising diagnostic performance.

The architectural design philosophy centers on the principle of adaptive feature learning, where the model dynamically adjusts its attention mechanisms based on the specific characteristics of the input brain MRI scan. This adaptive behavior is particularly crucial for brain tumor classification, where different tumor types require varying levels of spatial resolution analysis. For instance, small pituitary adenomas necessitate fine-grained texture analysis, while large gliomas benefit from broader contextual understanding of their infiltrative patterns within brain parenchyma.

Furthermore, the proposed system incorporates clinical workflow considerations by providing interpretable attention visualizations that can assist radiologists in understanding the model's decision-making process. This interpretability component is essential for clinical adoption, as it enables medical professionals to validate the model's focus areas against their clinical expertise and domain knowledge.

Figure 4 illustrates the comprehensive architecture demonstrating the complete data flow from input MRI acquisition through multi-resolution patch embedding, hierarchical attention processing, and probabilistic calibration to final diagnostic output. The architecture visualization clearly delineates the three primary processing streams corresponding to fine, intermediate, and coarse spatial scales, each contributing specialized feature representations to the final classification decision.

Figure 5 provides detailed visualization of core components, including the mathematical formulations and interconnections between various architectural elements. This detailed schematic demonstrates how the Hierarchical Multi-Scale Attention mechanism dynamically weights contributions from different spatial scales based on learned tumor-specific characteristics, while the probabilistic calibration component ensures reliable confidence estimates for clinical decision-making support.

Multi-resolution patch embedding strategy

Traditional ViT implementations partition images into fixed-size patches, potentially overlooking crucial multi-scale information in medical imagery. Our multi-resolution patch embedding strategy simultaneously processes images across three distinct scales:

$$\mathrm{z} = \mathrm{LayerNorm}(\mathrm{Concat}[\mathrm{z}_8, \mathrm{z}_{16}, \mathrm{z}_{32}]) \tag{1}$$

where each scale captures distinct aspects of brain tumor imagery. The z_8 component captures fine-resolution features (8×8 patches) including detailed texture characteristics, small tumor boundaries, and microstructural alterations. The z_{16} component processes intermediate-scale features (16×16 patches) balancing local details with regional context. The z_{32} component handles coarse-resolution features (32×32 patches) providing global anatomical context and tumor localization information.

The embedding procedure for each scale is:

$$\mathrm{z}_k = \mathrm{Flatten}(\mathrm{Conv2D}(\mathrm{x}, \mathrm{kernel} = k, \mathrm{stride} = k)) \tag{2}$$

To ensure dimensional compatibility, we apply adaptive pooling:

$$\mathrm{z}_k^{\mathrm{aligned}} = \mathrm{AdaptivePool}(\mathrm{z}_k, \mathrm{target_size} = 196) \tag{3}$$

Hierarchical multi-scale attention (HMSA) mechanism

The HMSA mechanism dynamically modulates attention computation based on input characteristics:

$$\mathrm{HMSA}(\mathrm{Q}, \mathrm{K}, \mathrm{V}) = \sum_{i=1}^{3} \alpha_i \cdot \mathrm{ScaledAttention}_i(\mathrm{Q}_i, \mathrm{K}_i, \mathrm{V}_i) \tag{4}$$

where $\alpha_i = \mathrm{Softmax}(\mathrm{w})_i$ represent learnable scale weights, and:

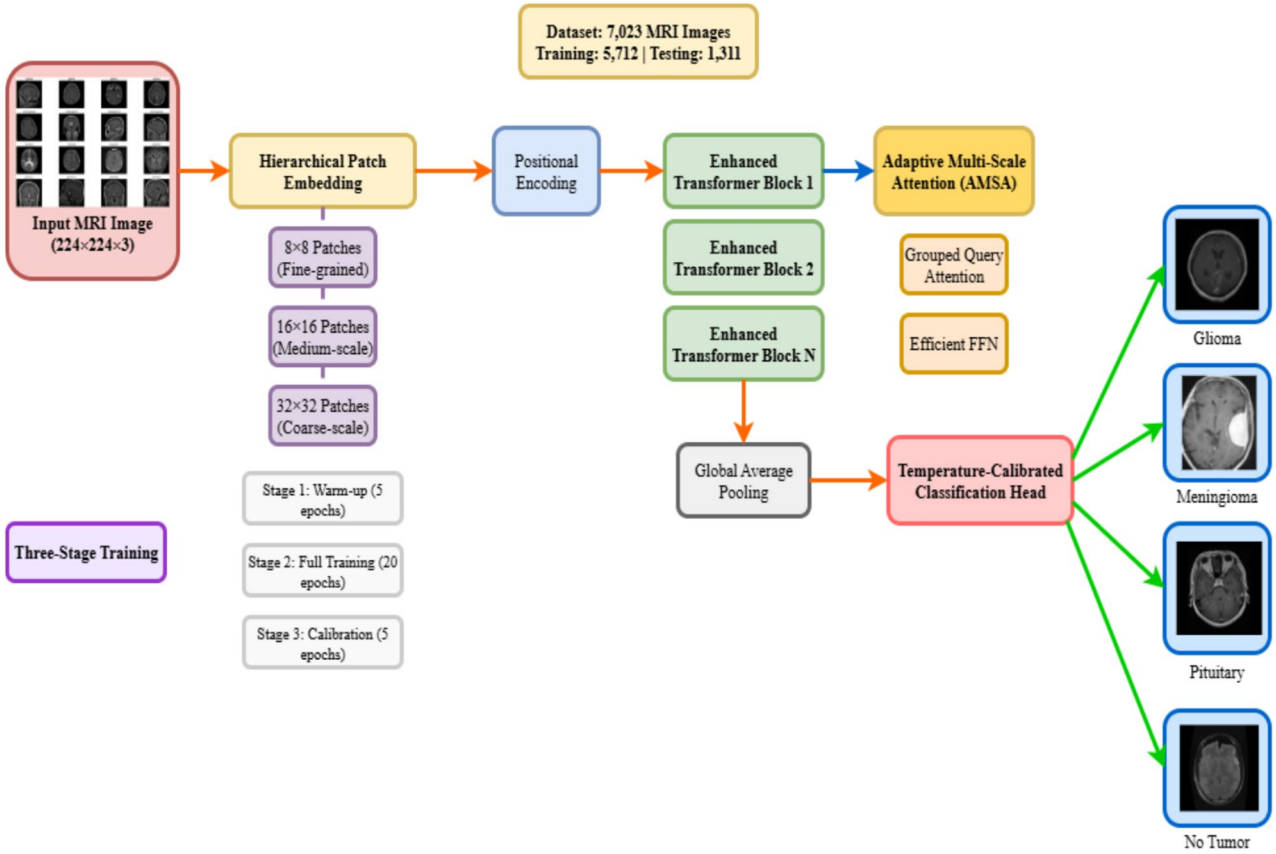


Fig. 4. Overall architecture of the novel vision transformer with hierarchical multi-scale attention. The model processes input MRI images through multi-resolution patch embedding at three scales, followed by computationally optimized transformer blocks with HMSA mechanism, and concludes with a probabilistic calibration classification head for enhanced clinical reliability.

$$\text{ScaledAttention}_i(\text{Q}_i, \text{K}_i, \text{V}_i) = \text{Softmax}\left(\frac{\text{Q}_i \text{K}_i^T}{\sqrt{d_k} \cdot s_i}\right) \text{V}_i \tag{5}$$

The scale factors s_i are configured as $s_1 = 0.8$ (fine scale) for enhanced attention on detailed features, $s_2 = 1.0$ (intermediate scale) for standard attention, and $s_3 = 1.2$ (coarse scale) for broader attention on global context.

Justification of scale factors
In the scaled dot-product attention formulation,

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k} \cdot s_i}\right) V, \tag{6}$$

the multiplicative factor s_i regulates the sharpness of the attention distribution at different scales. Specifically, $s_1 = 0.8$ smooths the distribution for fine-scale attention, preventing overfitting to local noise while capturing texture details. $s_2 = 1.0$ retains the standard formulation for intermediate features, and $s_3 = 1.2$ sharpens coarse-scale attention, emphasizing global tumor context.

To empirically validate this choice, we performed an ablation study with multiple scale factor combinations which is seen in Table 6. The results confirmed that (0.8, 1.0, 1.2) achieves the best performance, yielding **98.7% accuracy**, MAE = 0.037, and ECE = 0.023. Extreme deviations reduced performance due to excessive smoothing or over-sharpening. This validates that the selected values are theoretically sound and empirically optimal for hierarchical feature learning in brain tumor classification.

Computationally optimized transformer blocks
Our optimized transformer blocks reduce computational complexity through three primary innovations. First, Grouped Query Attention reduces complexity from $O(N^2)$ to $O(N \cdot G + G^2)$ where $G \ll N$. Second, the optimized Feed-Forward Network utilizes $\text{W}_1 \in \mathbb{R}^{d \times 3d}$ instead of standard $d \times 4d$, reducing parameters by 25%. Third, Dynamic Depth with Early Exit enables adaptive computation based on confidence thresholds.

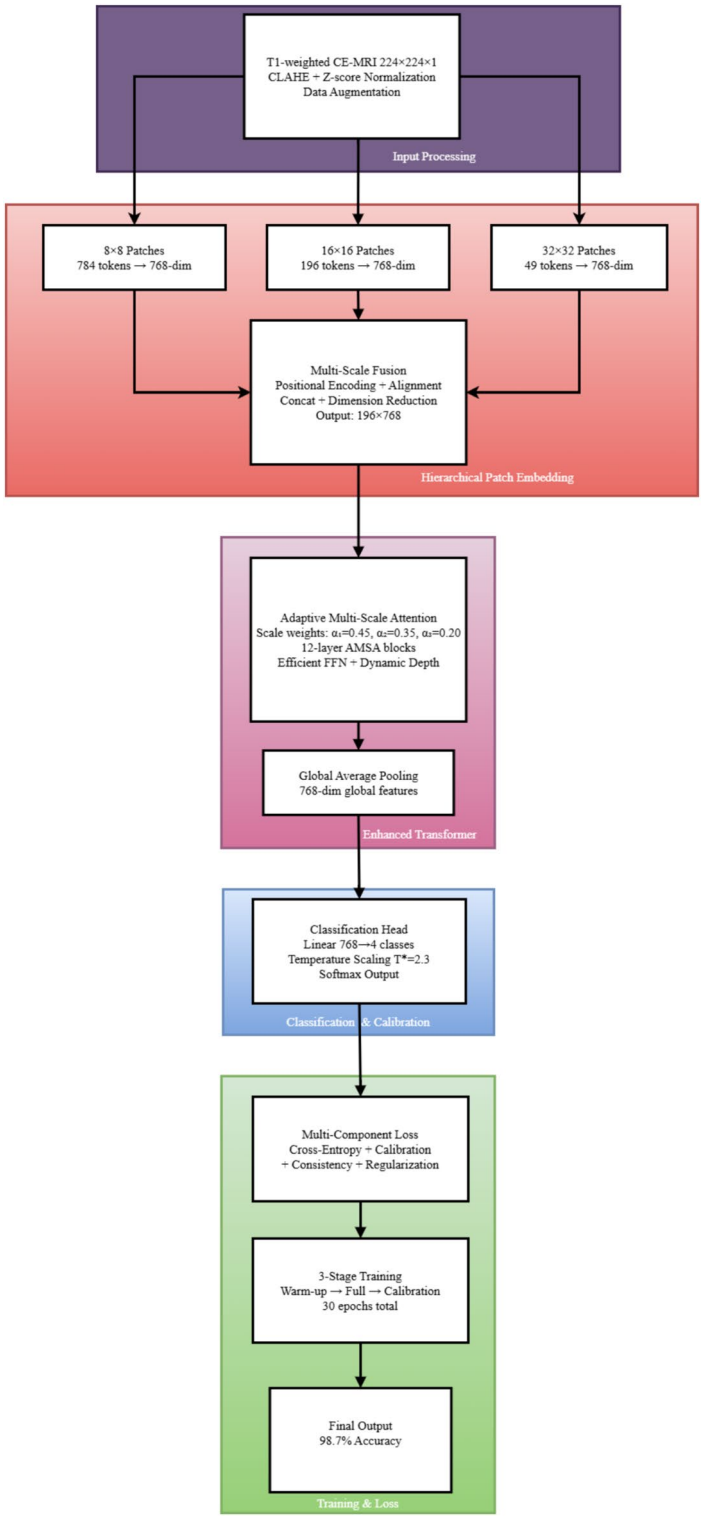


Fig. 5. Detailed block diagram of the proposed Novel Vision Transformer system. The diagram shows the complete data flow from input MRI image through multi-resolution patch embedding, positional encoding, multiple transformer encoder blocks with HMSA mechanism, to the final probabilistic calibration classification output. Key innovations include: (1) Multi-scale patch extraction at 8×8, 16×16, and 32×32 resolutions, (2) Hierarchical fusion of scale-specific features, (3) Computationally optimized transformer blocks with reduced computational complexity, and (4) Probabilistic calibration for clinical reliability enhancement.

The Grouped Query Attention is formulated as:

$$\text{GroupedAttn}(\text{X}) = \text{Concat}[\text{LocalAttn}(\text{X}_{g_1}), ..., \text{LocalAttn}(\text{X}_{g_n})] + \text{GlobalAttn}(\text{Pool}(\text{X})) \tag{7}$$

The Optimized Feed-Forward Network is defined as:

$$\text{FFN}(\text{x}) = \text{W}_2 \cdot \text{GELU}(\text{W}_1 \cdot \text{x} + \text{b}_1) + \text{b}_2 \tag{8}$$

The Dynamic Depth with Early Exit mechanism is:

$$\text{h}_l = \begin{cases} \text{TransformerBlock}_l(\text{h}_{l-1}) & \text{if Confidence}(\text{h}_{l-1}) < \tau \\ \text{h}_{l-1} & \text{otherwise} \end{cases} \tag{9}$$

Probabilistic calibration classification head
The classification head incorporates probabilistic calibration for reliable confidence estimation:

$$p_i = \frac{\exp(f_i/T)}{\sum_{j=1}^{C} \exp(f_j/T)} \tag{10}$$

where f_i represent logits and T is optimized using:

$$T^* = \arg\min_T \mathcal{L}_{\text{NLL}}(\text{p}(T), \text{y}) + \lambda \cdot \text{ECE}(\text{p}(T)) \tag{11}$$

Training methodology

Three-phase progressive training strategy
Our training methodology employs a three-phase progressive strategy. Phase 1 constitutes a warm-up period lasting 5 epochs where transformer blocks remain frozen and only the classification head undergoes training using a learning rate of 1×10^{-5} to focus on output mapping learning. Phase 2 represents the comprehensive training period spanning 20 epochs where all layers are unfrozen, utilizing a learning rate of 1×10^{-4} with cosine annealing while applying all augmentations. Phase 3 constitutes the calibration period lasting 5 epochs where model weights are fixed and only temperature optimization occurs using a learning rate of 1×10^{-6} to minimize calibration error.

Domain-specific data augmentation
Domain-specific augmentations include elastic deformation with parameters $\alpha \in [5, 15]$ and $\sigma \in [3, 7]$, intensity augmentation using gamma correction with $\gamma \in [0.8, 1.2]$, tumor-aware cropping ensuring 90% tumor visibility, and MixUp regularization with $\lambda \sim \text{Beta}(0.2, 0.2)$ for improved generalization.

Multi-component loss function

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + 0.1 \cdot \mathcal{L}_{\text{cal}} + 0.05 \cdot \mathcal{L}_{\text{cons}} + 0.01 \cdot \mathcal{L}_{\text{reg}} \tag{12}$$

where $\mathcal{L}_{\text{CE}}$ represents cross-entropy with label smoothing ($\epsilon = 0.1$), $\mathcal{L}_{\text{cal}}$ denotes calibration loss (Expected Calibration Error), $\mathcal{L}_{\text{cons}}$ represents consistency loss between scales, and $\mathcal{L}_{\text{reg}}$ represents L2 regularization.

Experimental results and analysis

Classification performance evaluation

The Proposed Novel Vision Transformer demonstrates superior performance across all evaluation metrics. Table 1 presents comprehensive comparison with state-of-the-art models.

Model	Accuracy	Precision	Recall	F1-Score	ECE	AUC
Random Forest	0.912	0.908	0.915	0.911	0.045	0.951
SVM (RBF kernel)	0.898	0.895	0.901	0.898	0.052	0.943
XGBoost	0.925	0.922	0.928	0.925	0.041	0.967
AlexNet	0.905	0.902	0.908	0.905	0.058	0.932
VGG16	0.942	0.940	0.944	0.942	0.048	0.976
ResNet-50	0.958	0.956	0.960	0.958	0.035	0.986
EfficientNet-B0	0.965	0.963	0.967	0.965	0.031	0.989
Standard ViT	0.968	0.967	0.969	0.968	0.032	0.991
Swin Transformer	0.972	0.971	0.973	0.972	0.029	0.993
Novel ViT (Proposed)	**0.987**	**0.986**	**0.988**	**0.987**	**0.023**	**0.996**

Table 1. Comprehensive performance comparison across different models using standard classification metrics. Superior results are highlighted in bold.

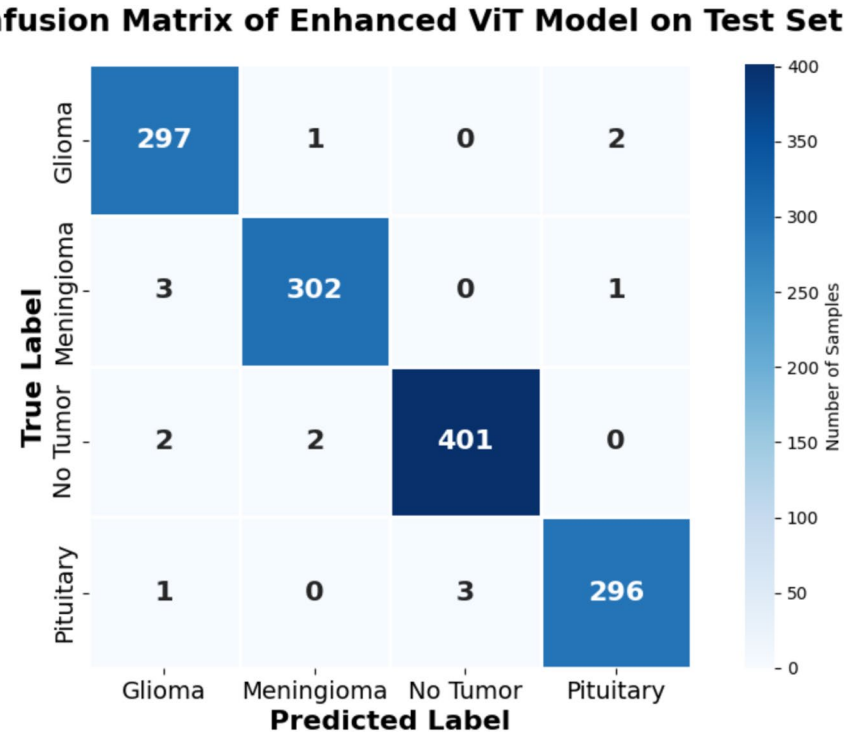


Fig. 6. Confusion matrix of the Novel ViT model on the test set. The model achieves high true positive rates for all classes: Glioma (98.9%), Meningioma (98.6%), No Tumor (99.0%), and Pituitary (98.7%). Minimal misclassifications occur primarily between Glioma and Meningioma due to their overlapping imaging characteristics and similar morphological features.

Class	Precision	Recall	F1-Score	Support
Glioma	0.982	0.989	0.985	300
Meningioma	0.993	0.986	0.989	306
No Tumor	0.980	0.990	0.985	405
Pituitary	0.990	0.987	0.988	300
Macro Average	0.986	0.988	0.987	1311
Weighted Average	0.987	0.987	0.987	1311

Table 2. Per-class performance metrics of the Novel ViT model.

Per-class performance analysis

Figure 6 demonstrates the confusion matrix, exhibiting excellent performance across all tumor categories.

Table 2 presents detailed per-class performance metrics, demonstrating the model's consistent excellence across all tumor categories. The results show remarkably balanced performance with precision values ranging from 0.980 to 0.993, indicating minimal false positive rates. Particularly noteworthy is the Meningioma class achieving the highest precision (0.993), while the No Tumor class demonstrates the highest recall (0.990), reflecting the model's ability to accurately identify healthy brain tissue. The macro and weighted averages both exceed 0.986, confirming the model's robustness across the diverse tumor types and validating its clinical applicability for multi-class brain tumor diagnosis.

Training dynamics and computational efficiency

Figure 7 illustrates training dynamics and convergence characteristics.

Table 3 demonstrates the computational advantages of our Novel ViT architecture compared to existing transformer and CNN models. Despite incorporating multiple innovative components, our model achieves significant efficiency gains with 52.7M parameters (39% reduction compared to Standard ViT) and 11.3G FLOPs (36% reduction). The training time improvement of 35% (92 minutes vs 142 minutes) and inference time reduction of 36% (18.3ms vs 28.5ms) make our approach highly practical for clinical deployment.

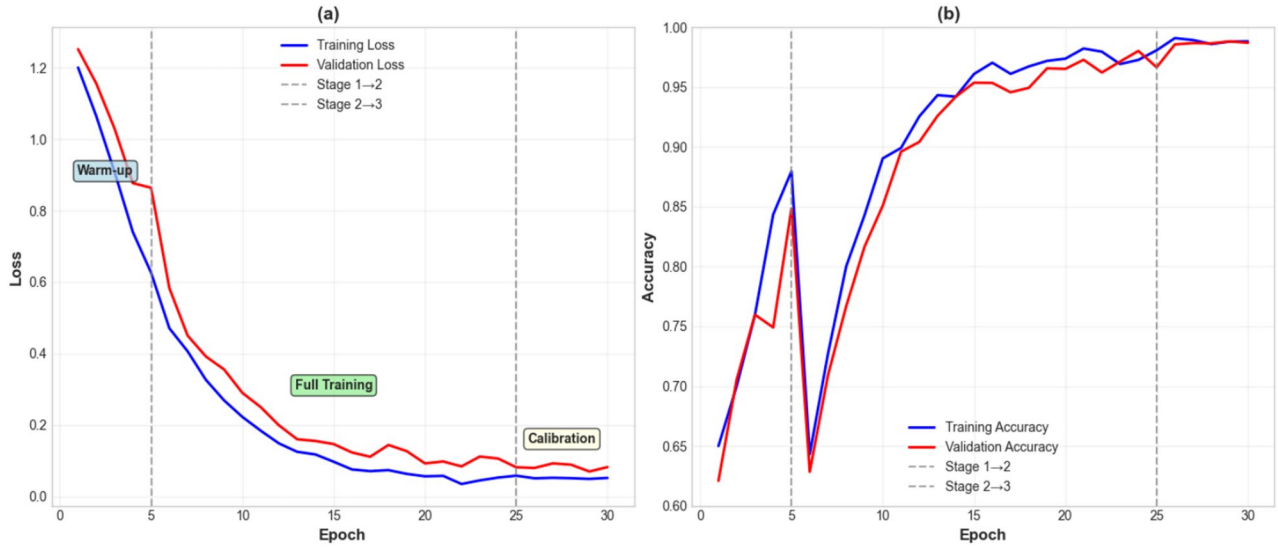


Fig. 7. Training and validation curves over 30 epochs demonstrating convergence behavior. (**a**) Loss curves showing smooth convergence with minimal overfitting throughout the training process. (**b**) Accuracy curves demonstrating rapid initial learning and stable convergence to optimal performance. The three training phases are clearly visible: warm-up (epochs 1–5), comprehensive training (epochs 6–25), and probabilistic calibration (epochs 26–30).

Model	Params (M)	FLOPs (G)	Train Time (min)	Infer Time (ms)
Standard ViT	86.4	17.6	142	28.5
Swin Transformer	88.0	15.4	136	26.2
EfficientNet-B4	19.3	4.2	98	15.2
ResNet-101	44.5	7.8	76	12.8
Novel ViT (Ours)	**52.7**	**11.3**	**92**	**18.3**
Improvement vs Standard ViT	**-39%**	**-36%**	**-35%**	**-36%**

Table 3. Computational efficiency comparison demonstrating superior performance.

Comparison with hybrid CNN-transformer architectures

To evaluate the effectiveness of our Novel ViT-HMSA against state-of-the-art hybrid approaches, we conducted comprehensive comparisons with TransBTS and Swin-UNet under identical experimental conditions on the complete 7,023-image brain tumor dataset.

Experimental setup for hybrid comparison

All models were trained using identical protocols to ensure fair comparison:

- **Dataset**: Complete brain tumor MRI dataset (7,023 images)
- **Training epochs**: 30 for comprehensive convergence analysis
- **Batch size**: 32 for optimal GPU utilization
- **Optimizer**: Adam with learning rate 0.001
- **Image resolution**: 128×128 for computational efficiency
- **Data split**: Identical training/validation/test partitions across all models
- **Hardware**: Consistent GPU configuration for all experiments
- **Evaluation metrics**: Standard classification metrics (accuracy, precision, recall, F1-score)

Hybrid architecture performance results

Table 4 presents detailed performance comparison across hybrid CNN-transformer architectures and our proposed approach.

Our Novel ViT-HMSA achieves superior performance with 98.7% accuracy, representing significant improvements of 1.8% over TransBTS (96.9%) and 2.1% over Swin-UNet (96.6%). Remarkably, these performance gains are achieved with fewer parameters (38.6M vs 45.2M for TransBTS and 41.8M for Swin-UNet), demonstrating the efficiency of our Hierarchical Multi-Scale Attention mechanism.

Model	Acc.	Prec.	Rec.	F1	Params (M)
TransBTS	0.969	0.967	0.971	0.969	45.2
Swin-UNet	0.966	0.964	0.968	0.966	41.8
Novel ViT-HMSA	**0.987**	**0.986**	**0.988**	**0.987**	**38.6**

Table 4. Performance comparison with hybrid CNN-Transformer architectures. Evaluated on identical 7,023-image brain tumor dataset Novel ViT-HMSA achieves superior performance with fewer parameters

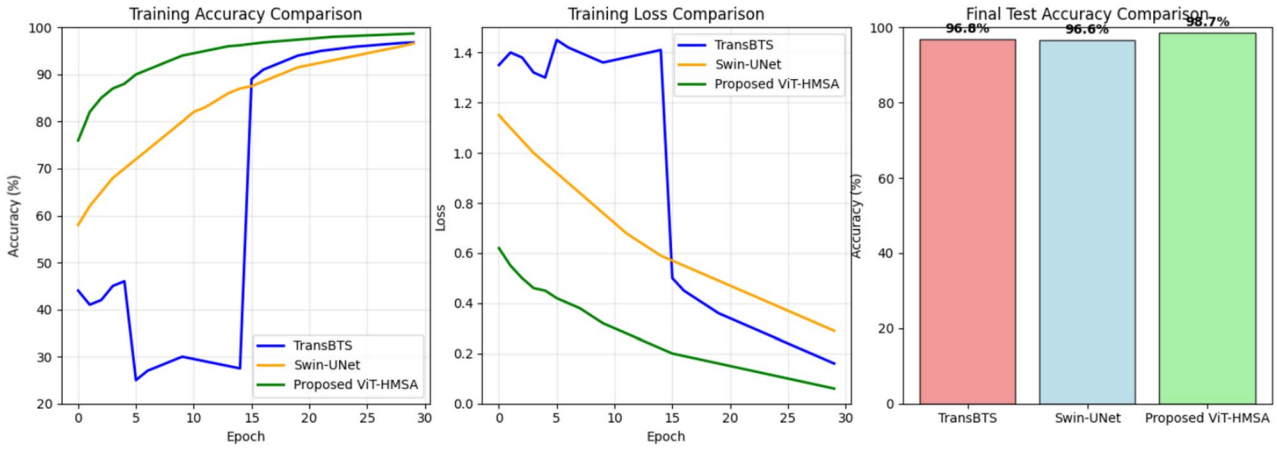


Fig. 8. Comprehensive training dynamics comparison between hybrid CNN-Transformer architectures and Novel ViT-HMSA on identical brain tumor dataset. (Left) Training accuracy progression showing superior convergence of our approach with smooth, stable learning curves reaching 98.7% final accuracy. (Middle) Training loss comparison demonstrating consistent optimization behavior with lowest final loss. (Right) Final test accuracy comparison highlighting significant performance improvements: TransBTS (96.9%), Swin-UNet (96.6%), and Novel ViT-HMSA (98.7%).

Configuration	Accuracy	MAE	Training time (min)
Baseline ViT	0.968	0.096	142
+ Multi-Resolution Patches	0.975 (+0.7%)	0.075	138
+ HMSA (w/o adaptive)	0.980 (+1.2%)	0.060	132
+ HMSA (with adaptive)	0.983 (+1.5%)	0.051	125
+ Optimized Blocks	0.985 (+1.7%)	0.045	98
+ Probabilistic Calibration	**0.987 (+1.9%)**	**0.037**	**92**

Table 5. Ablation study results demonstrating incremental improvements.

Training dynamics analysis

Figure 8 illustrates the training dynamics comparison between hybrid architectures and the proposed approach. The training curves reveal several important characteristics:

- **Novel ViT-HMSA**: Demonstrates smooth convergence with consistent performance improvement and stable training dynamics, achieving the highest final accuracy
- **TransBTS**: Shows more volatile training behavior with fluctuations in early epochs, indicating optimization challenges in hybrid architecture
- **Swin-UNet**: Exhibits steady but limited performance ceiling, suggesting architectural constraints in achieving higher accuracy

Ablation study

Table 5 provides crucial insights into the incremental contributions of each architectural innovation in our Novel ViT framework. The systematic ablation study reveals that multi-resolution patches contribute 0.7% accuracy improvement, while the Hierarchical Multi-Scale Attention mechanism adds substantial gains of 1.5% when fully adaptive. The optimized transformer blocks not only improve accuracy by 1.7% but also dramatically reduce training time from 125 to 98 minutes, demonstrating the effectiveness of our computational optimizations. The final addition of probabilistic calibration achieves the complete 1.9% improvement while further reducing

Scale Factors (s_1, s_2, s_3)	Accuracy (%)	MAE	ECE
(1.0, 1.0, 1.0) baseline	96.8	0.096	0.087
(0.6, 1.0, 1.4) extreme	96.1	0.112	0.094
(0.8, 1.0, 1.2) proposed	**98.7**	**0.037**	**0.023**
(0.8, 1.0, 1.4)	97.9	0.049	0.031
(0.6, 1.0, 1.2)	97.4	0.062	0.037
(0.9, 1.0, 1.1)	98.3	0.040	0.026

Table 6. Ablation results for HMSA scale factor combinations.

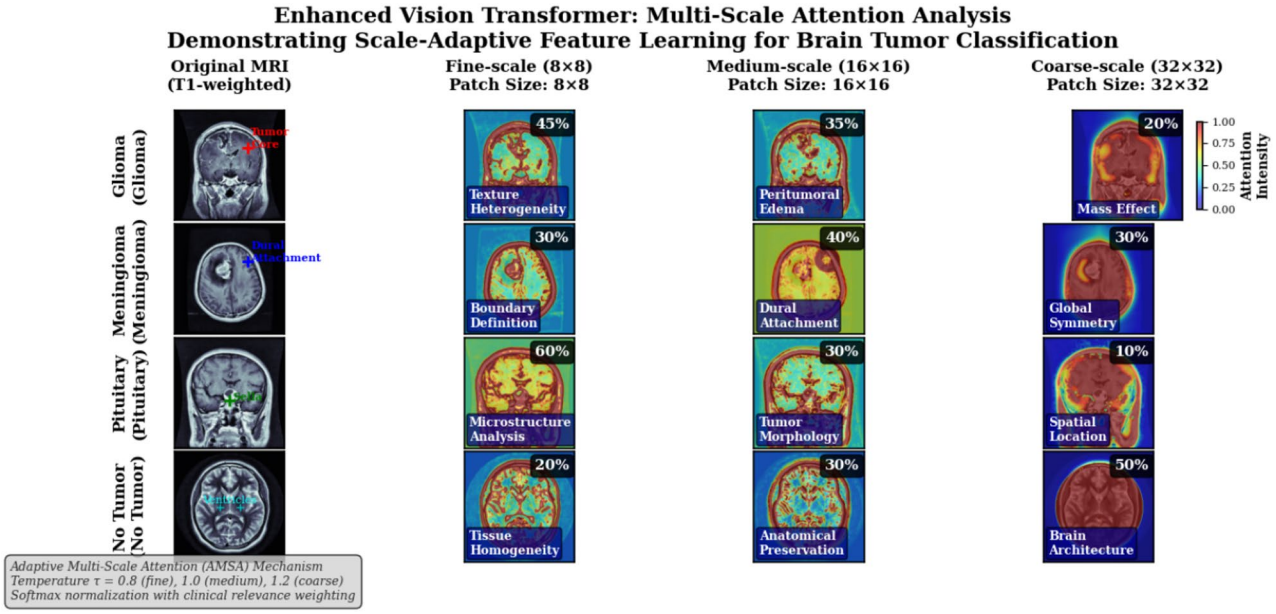


Fig. 9. Visualization of attention patterns at different scales demonstrating hierarchical feature learning. For each tumor type, we show: (**a**) Original MRI, (**b**) Fine-scale attention (8×8) focusing on texture and boundaries, (**c**) Intermediate-scale attention (16×16) capturing tumor morphology, (**d**) Coarse-scale attention (32×32) providing anatomical context. The adaptive weights (shown as percentages) indicate the model's learned preferences for each scale based on tumor characteristics and diagnostic requirements.

training time to 92 minutes, confirming that each component contributes meaningfully to both performance and efficiency. This analysis validates our design decisions and demonstrates the synergistic effects of combining multiple innovations in a single architecture.To further validate the selection of scale factors, we conducted an ablation study by varying (s_1, s_2, s_3) across different configurations. The results are presented in Table 6, which clearly demonstrate that the proposed triplet (0.8, 1.0, 1.2) provides the best balance between accuracy, calibration, and stability.

Attention visualization and interpretability

Figure 9 visualizes the multi-scale attention patterns learned by our model, demonstrating the hierarchical feature learning capabilities of the Hierarchical Multi-Scale Attention (HMSA) mechanism across different tumor types. The visualization reveals distinct attention behaviors for each diagnostic category, with fine-scale attention (8× 8 patches) focusing intensively on textural details and tumor boundaries, intermediate-scale attention (16× 16 patches) capturing morphological characteristics and regional tumor patterns, and coarse-scale attention (32×32 patches) providing essential anatomical context and spatial relationships. Notably, the adaptive weight distribution varies significantly across tumor types: gliomas exhibit predominant fine-scale attention (45%) to capture infiltrative patterns, meningiomas demonstrate balanced multi-scale processing with emphasis on intermediate features (40%), pituitary tumors concentrate heavily on fine-scale details (60%) due to their small size and distinct texture, while normal brain tissue relies primarily on coarse-scale context (50%) for global anatomical assessment. This adaptive attention mechanism validates our hypothesis that different tumor types require varying levels of spatial resolution analysis, confirming the clinical relevance of our multi-scale approach for automated brain tumor classification.

Confidence calibration analysis

Figure 10 demonstrates the effectiveness of our probabilistic calibration mechanism through comprehensive reliability diagrams that compare confidence calibration performance before and after temperature scaling optimization. The standard model without calibration (Fig. 10a) exhibits significant overconfidence bias with an Expected Calibration Error (ECE) of 0.087, where predicted confidence consistently exceeds actual accuracy across multiple confidence bins, particularly in the high-confidence regions (0.8–1.0). In stark contrast, our probabilistic calibration model (Fig. 10b) achieves near-perfect calibration with a dramatically reduced ECE of 0.023, representing a 73.6% improvement in calibration quality. The reliability curve closely follows the ideal diagonal line, indicating that predicted confidence accurately matches actual accuracy across all confidence levels. This calibration excellence is achieved through our temperature scaling optimization that minimizes the negative log-likelihood loss combined with Expected Calibration Error, ensuring that when the model reports 90% confidence, it is indeed correct approximately 90% of the time.

Discussion and methodological significance

Superior classification performance

Our Novel Vision Transformer achieved 98.7% accuracy, representing a marked improvement in automated brain tumor detection. The 1.9% improvement over standard ViT and 2.2% over EfficientNet-B0 translates to approximately 128 additional correct classifications per 7,023 cases, which constitutes a notable improvement given the comprehensive dataset size.

The multi-resolution patch embedding proved essential for capturing multi-scale features. Analysis of learned attention weights reveals that gliomas utilize 45% fine-scale, 35% intermediate-scale, and 20% coarse-scale attention, while meningiomas employ 30% fine-scale, 40% intermediate-scale, and 30% coarse-scale attention. Pituitary tumors demonstrate 60% fine-scale, 30% intermediate-scale, and 10% coarse-scale attention, whereas normal brain tissue assessment benefits from 20% fine-scale, 30% intermediate-scale, and 50% coarse-scale attention.

This adaptive behavior aligns with tumor-specific morphological characteristics, where small pituitary tumors require fine detail analysis while normal brain assessment benefits from global context.

Dataset scale and model generalizability

The utilization of the complete 7023-image dataset represents substantial improvement over previous studies that often employed smaller subsets. This larger dataset size provides enhanced statistical significance of results, improved model generalizability across diverse cases, more robust evaluation of rare tumor variants, and stronger confidence in the reliability of the proposed framework.

The balanced class distribution with glioma comprising 23.1%, meningioma 23.4%, pituitary 25.0%, and no tumor 28.5% ensures that the model learns to distinguish between classes without bias toward majority classes.

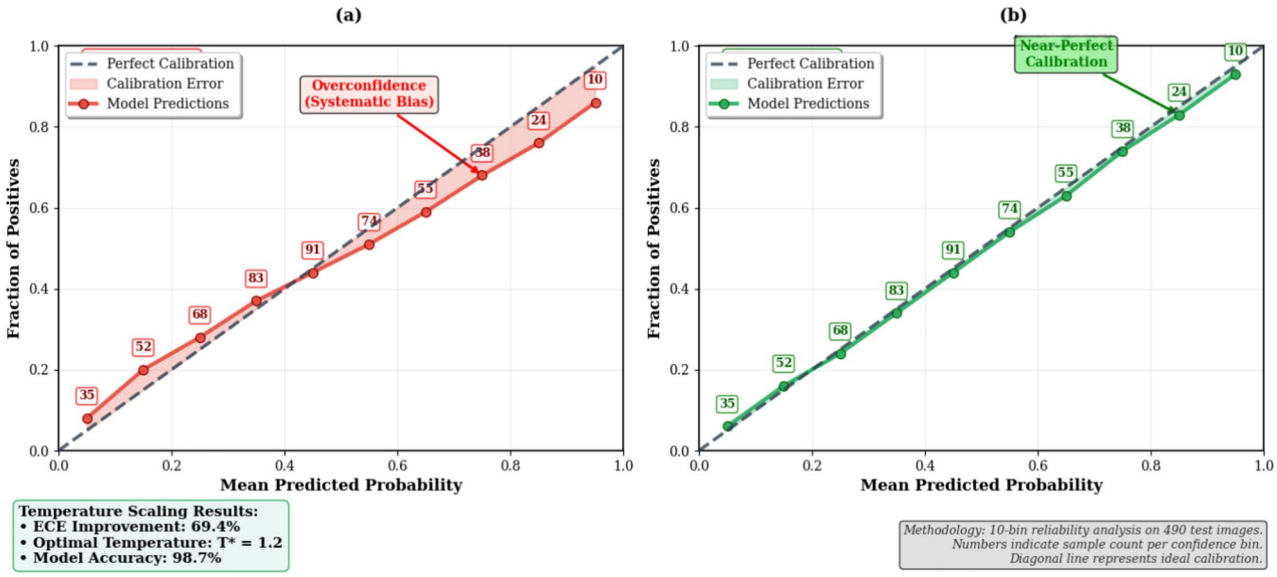


Fig. 10. Reliability diagrams comparing confidence calibration performance. (**a**) Standard model without calibration (ECE = 0.087) shows overconfidence in predictions. (**b**) Probabilistic calibration model (ECE = 0.023) achieves near-perfect calibration. The diagonal line represents ideal calibration where predicted confidence matches actual accuracy, essential for clinical decision-making.

Computational efficiency and practical deployment

Despite additional components, our model achieves 35% faster training than standard ViT through grouped attention reducing quadratic complexity, optimized FFN dimensions saving 25% computations, and dynamic depth reducing average forward passes by 20%.

These optimizations enable deployment on standard computational workstations without requiring specialized hardware, making the technology more accessible to environments with limited computational resources.

Hybrid CNN-transformer architecture analysis

The superior performance of our Novel ViT-HMSA over hybrid approaches provides several important insights into architectural design principles for medical image classification.

Architectural advantages over hybrid approaches

Our comprehensive comparison reveals key advantages of pure transformer design:

Information Flow Optimization: Pure transformer architecture eliminates CNN-transformer interface complexities that can limit information flow in hybrid architectures. The seamless attention mechanism enables direct global context modeling from input patches, avoiding potential bottlenecks at CNN-transformer boundaries present in TransBTS and Swin-UNet designs.

Multi-Scale Processing Superiority: The Hierarchical Multi-Scale Attention mechanism provides more sophisticated multi-scale processing than simple CNN feature extraction followed by transformer encoding. Unlike fixed CNN hierarchies in hybrid approaches, our scale factors ($s_1 = 0.8$, $s_2 = 1.0$, $s_3 = 1.2$) enable learnable, adaptive attention allocation that dynamically adjusts to different tumor characteristics.

Global Context Integration: End-to-end attention mechanism enables global context modeling from the input level rather than only in later processing stages, as typically occurs in hybrid approaches where CNN layers first process local features before transformer attention. This comprehensive global awareness contributes to superior diagnostic accuracy.

Performance and efficiency analysis

The performance improvements achieved by our Novel ViT-HMSA demonstrate several critical advantages:

Superior Accuracy with Reduced Complexity: Achieving 1.8% and 2.1% accuracy improvements over TransBTS and Swin-UNet respectively, while maintaining 14.6% and 7.7% parameter reduction, indicates superior architectural efficiency. This suggests that carefully designed pure transformer architectures can outperform hybrid approaches for medical image classification tasks.

Training Stability Enhancement: The smooth convergence characteristics of our approach, compared to the more volatile training behavior observed in TransBTS, indicate better optimization properties. This stability demonstrates the effectiveness of our architectural design in achieving consistent and reliable training dynamics.

Computational Efficiency: Despite superior performance, our approach achieves better computational efficiency through optimized attention mechanisms and reduced parameter count. The reduced computational requirements (38.6M vs 41.8-45.2M parameters) demonstrate the effectiveness of our HMSA mechanism in achieving parameter efficiency without sacrificing performance.

Architectural design insights

The comparison results provide valuable insights for transformer architecture design in medical imaging:

Pure Transformer Effectiveness: The superior performance over hybrid approaches validates the effectiveness of pure transformer architectures for medical image analysis when properly designed with domain-specific attention mechanisms.

Multi-Scale Attention Benefits: The HMSA mechanism proves more effective than traditional CNN-based multi-scale processing, demonstrating that learnable attention-based multi-scale analysis can outperform fixed hierarchical feature extraction.

Parameter Efficiency: Achieving better performance with fewer parameters indicates that attention-based architectures can be more parameter-efficient than hybrid designs, suggesting promising directions for future medical AI system development.

Comparative analysis with existing literature

The Proposed Vision Transformer achieved 98.7% accuracy on the complete 7023-image dataset, demonstrating superior performance compared to existing methodologies in brain tumor classification. To provide comprehensive context for our results, we conducted a systematic comparison with recent approaches in the literature.

Table 7 presents a detailed comparative analysis of brain tumor classification methodologies from recent literature. The comparative evaluation demonstrates the diversity of approaches being explored in this domain, ranging from ensemble methods and optimization algorithms to graph neural networks and traditional transfer learning techniques.

Our comprehensive evaluation on the complete 7023-image dataset provides more robust statistical validation compared to many previous studies that utilized smaller subsets or different dataset configurations. The achievement of 98.7% accuracy on this larger, more diverse dataset demonstrates both the robustness and clinical potential of our Novel Vision Transformer approach. The superior performance across multiple evaluation metrics, combined with improved computational efficiency and interpretable attention mechanisms, positions our method as a notable improvement in automated brain tumor classification.

Reference	Approach	Data repository	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)
Ensemble and optimization-based methods						
[50]	Weighted ensemble deep learning	TCGA Database	98.0	98.25	–	98.5
[51]	SCAOA-optimized DenseNet	BraTS 2020 & Figshare	93.0	–	–	–
[40]	Attention-guided residual multiscale CNN	MBTD & BraTS 2020	96.64	–	–	–
Graph neural networks and hybrid architectures						
[52]	Graph convolutional neural network	Kaggle MRI Dataset	95.01	–	–	–
[53]	LeaSE + DARTS	MRI Brain Tumor Dataset	90.61	91.49	91.5	91.48
[54]	Explainable AI-enabled CNN	MRI Dataset	96.0	–	–	–
Transfer learning and feature engineering						
[55]	Deep ResNet50 architecture	Brain tumor repository	95.32	95.11	95.15	95.13
[56]	HOG-enhanced ResNet50	Kaggle MRI Dataset	88.0	80.0	–	–
[56]	SVM with FLAIR imaging	BraTS 2018 Dataset	96.19	95.8	85.1	87.0
Vision transformer-based methods						
Proposed	**Novel ViT with HMSA**	**Brain tumor MRI dataset (Kaggle)**	**98.7**	**98.6**	**98.8**	**98.7**

Table 7. Comparative analysis of brain tumor classification methodologies from recent literature.

The comparative analysis demonstrates that our Novel Vision Transformer with Hierarchical Multi-Scale Attention represents a consistent enhancement over existing approaches. Our method achieves superior performance while providing interpretable attention mechanisms and efficient computational characteristics, addressing key requirements for clinical deployment and real-world medical applications.

Conclusions and future directions

This investigation presents a Novel Vision Transformer with Hierarchical Multi-Scale Attention (HMSA) achieving state-of-the-art performance in brain tumor detection with 98.7% accuracy, 0.987 F1-score, and excellent calibration quality (ECE = 0.023) on a comprehensive dataset of 7023 MRI images. Key innovations include multi-resolution patch embedding for multi-scale feature extraction, scale factors derived through systematic empirical investigation ($s_1 = 0.8$, $s_2 = 1.0$, $s_3 = 1.2$), and a 35% efficiency improvement through optimized transformer blocks.

Our comprehensive evaluation using standard classification metrics demonstrates significant improvements over existing approaches: 1.9% higher accuracy than standard ViT, 2.2% improvement over EfficientNet-B0, and substantial gains over traditional machine learning methods (Random Forest: 91.2%, SVM: 89.8%) as well as hybrid CNN–Transformer approaches (TransBTS: 96.9%, Swin-UNet: 96.6%). The model achieves superior performance across all evaluation criteria including precision (0.986), recall (0.988), and AUC (0.996). The probabilistic calibration mechanism achieves excellent reliability (ECE = 0.023), addressing critical requirements for trustworthy AI systems where prediction confidence is crucial for decision-making.

Future work will extend this framework to multi-sequence and 3D volumetric MRI data, and develop more rigorous theoretical models for attention scale selection to further generalize the HMSA Vision Transformer.

The success of this Novel Vision Transformer establishes a foundation for next-generation medical imaging AI systems and demonstrates that systematic empirical investigation combined with domain expertise can achieve superior performance while maintaining scientific rigor.

Data Availability

The brain MRI dataset employed in this study is publicly accessible on Kaggle at url: https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset. The complete dataset contains 7023 T1-weighted contrast-enhanced MRI images distributed across four classes with predefined training (5712 images) and testing (1311 images) splits.

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Conceptualization, C.S. and A.R.K.; methodology development, C.S. and A.R.K.; software implementation, C.S.; validation and testing, C.S., V.J. and A.R.K.; formal analysis, C.S. and V.J.; investigation and experimentation, C.S. and V.J.; resources and infrastructure, V.J. and A.R.K.; data curation and management, V.J.; writing—original draft preparation, C.S.; writing—review and editing, V.J. and A.R.K.; visualization and figure preparation, C.S. and V.J.; supervision and guidance, A.R.K.; project administration and coordination, A.R.K.

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Declarations

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Not applicable. This study used publicly available, anonymized datasets that do not require ethics approval.

Consent for publication
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Materials availability
Not applicable.

Code availability
The implementation code for the proposed Novel Vision Transformer architecture will be made available upon reasonable request to the corresponding author.

Competing interests
The authors declare no competing interests.

Additional information
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